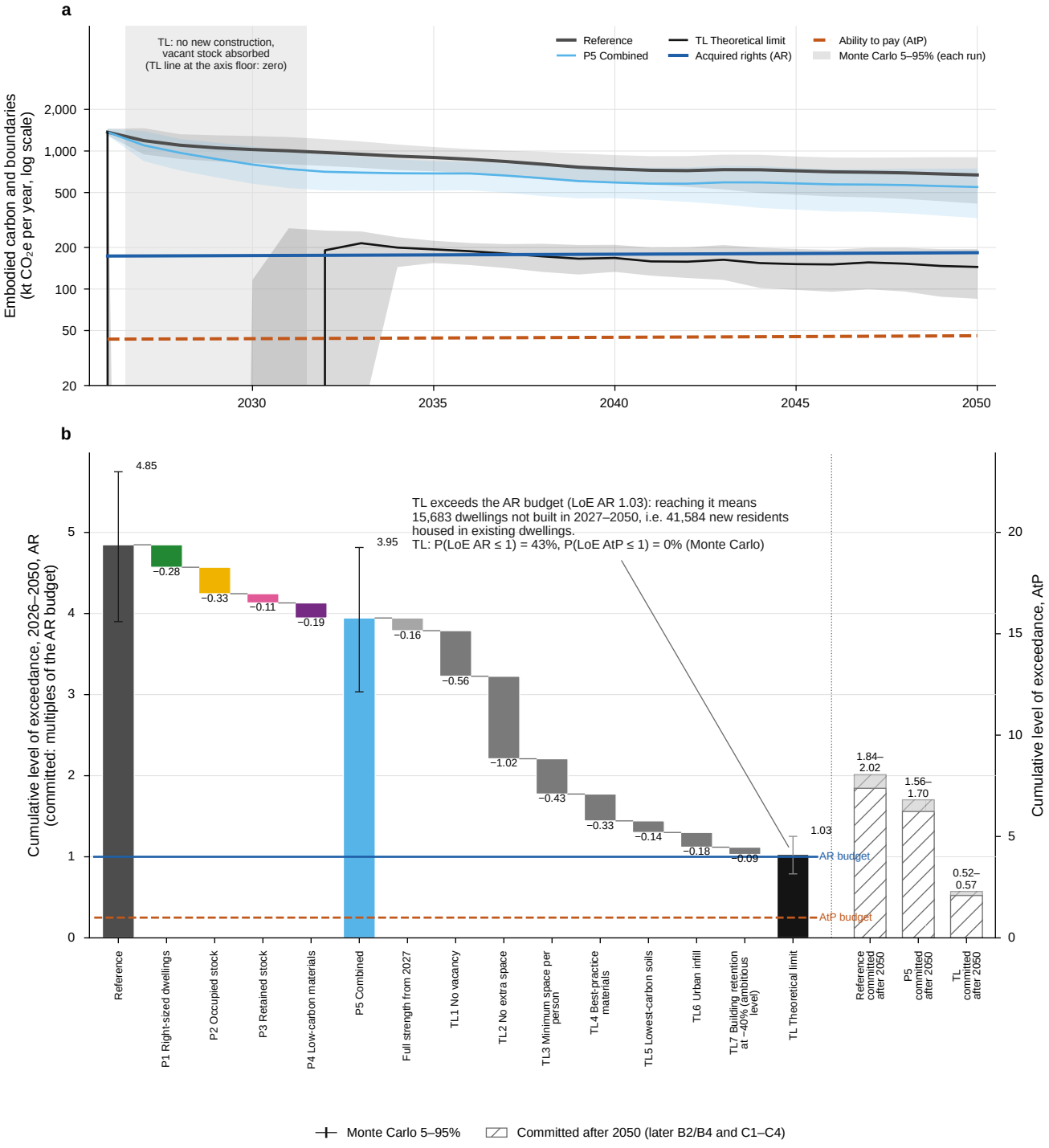
a
TL: no new construction, vacant stock absorbed (TL line at the axis floor: zero)
Reference
P5 Combined
TL Theoretical limit
Acquired rights (AR)
Ability to pay (AtP)
Monte Carlo 5–95% (each run)
Embodied carbon and boundaries (kt CO₂e per year, log scale)
2,000
1,000
500
200
100
50
20
2030
2035
2040
2045
2050
b
TL exceeds the AR budget (LoE AR 1.03): reaching it means 15,683 dwellings not built in 2027–2050, i.e. 41,584 new residents housed in existing dwellings.
TL: P(LoE AR ≤ 1) = 43%, P(LoE AtP ≤ 1) = 0% (Monte Carlo)
Cumulative level of exceedance, 2026–2050, AR (committed: multiples of the AR budget)
Cumulative level of exceedance, AtP
4.85
−0.28
−0.33
−0.11
−0.19
3.95
−0.16
−0.56
−1.02
−0.43
−0.33
−0.14
−0.18
−0.09
1.03
1.84–2.02
1.56–1.70
0.52–0.57
AR budget
AtP budget
Reference
P1 Right-sized dwellings
P2 Occupied stock
P3 Retained stock
P4 Low-carbon materials
P5 Combined
Full strength from 2027
TL1 No vacancy
TL2 No extra space
TL3 Minimum space per person
TL4 Best-practice materials
TL5 Lowest-carbon soils
TL6 Urban infill
TL7 Building retention at −40% (ambitious level)
TL Theoretical limit
Reference committed after 2050
P5 committed after 2050
TL committed after 2050
Monte Carlo 5–95%
Committed after 2050 (later B2/B4 and C1–C4)